

# Mann–Whitney U test for production and test code revisions

Two-sided Mann–Whitney U test comparing the number of revisions of **production** methods versus **test** methods per project. Each row reports the two-sided  $p$ -value, Cliff’s  $\delta$  effect size ( $d$ ), and its direction sign: + (43 project(s)) indicates that production methods have a higher revision count than test methods ( $\delta > 0$ ); – (66 project(s)) indicates the opposite, i.e. test methods are revised more frequently ( $\delta < 0$ ). Effect size bands follow Romano et al. thresholds for Cliff’s  $\delta$ : **N** negligible ( $|\delta| < 0.147$ ), **S** small ( $0.147 \leq |\delta| < 0.33$ ), **M** medium ( $0.33 \leq |\delta| < 0.474$ ), **L** large ( $|\delta| \geq 0.474$ ).

| project              | p-value | d-value | +/- | <i>Effect size band (Cliff’s <math>\delta</math>)</i> |   |   |   |
|----------------------|---------|---------|-----|-------------------------------------------------------|---|---|---|
|                      |         |         |     | N                                                     | S | M | L |
| Activiti             | 0.00    | -0.13   | -   | x                                                     |   |   |   |
| ant                  | 0.00    | 0.38    | +   |                                                       |   | x |   |
| AntennaPod           | 0.34    | 0.03    | +   | x                                                     |   |   |   |
| Apktool              | 0.00    | -0.21   | -   |                                                       | x |   |   |
| argouml              | 0.01    | 0.06    | +   | x                                                     |   |   |   |
| async-http-client    | 0.00    | 0.71    | +   |                                                       |   |   | x |
| atmosphere           | 0.00    | -0.12   | -   | x                                                     |   |   |   |
| auto                 | 0.13    | -0.04   | -   | x                                                     |   |   |   |
| baritone             | 0.92    | -0.02   | -   | x                                                     |   |   |   |
| byte-buddy           | 0.00    | 0.10    | +   | x                                                     |   |   |   |
| caffeine             | 0.00    | -0.40   | -   |                                                       |   | x |   |
| cassandra            | 0.01    | -0.02   | -   | x                                                     |   |   |   |
| checkstyle           | 0.00    | -0.38   | -   |                                                       |   | x |   |
| commons-io           | 0.09    | 0.03    | +   | x                                                     |   |   |   |
| commons-lang         | 0.00    | 0.34    | +   |                                                       |   | x |   |
| cucumber-jvm         | 0.00    | -0.32   | -   |                                                       | x |   |   |
| dagger               | 0.01    | 0.14    | +   | x                                                     |   |   |   |
| dbeaver              | 0.00    | 0.16    | +   |                                                       | x |   |   |
| docx4j               | 0.00    | 0.21    | +   |                                                       | x |   |   |
| dropwizard           | 0.00    | -0.38   | -   |                                                       |   | x |   |
| dubbo                | 0.00    | -0.33   | -   |                                                       | x |   |   |
| eclipse.jdt.core     | 0.00    | 0.12    | +   | x                                                     |   |   |   |
| elasticsearch        | 0.00    | -0.22   | -   |                                                       | x |   |   |
| error-prone          | 0.00    | -0.17   | -   |                                                       | x |   |   |
| Essentials           | 0.95    | 0.01    | +   | x                                                     |   |   |   |
| ExoPlayer            | 0.00    | -0.04   | -   | x                                                     |   |   |   |
| facebook-android-sdk | 0.01    | -0.49   | -   |                                                       |   |   | x |
| fastjson             | 0.00    | 0.54    | +   |                                                       |   |   | x |
| feign                | 0.00    | -0.53   | -   |                                                       |   |   | x |
| flink                | 0.00    | -0.34   | -   |                                                       |   | x |   |
| glide                | 0.44    | 0.01    | +   | x                                                     |   |   |   |
| grpc-java            | 0.00    | -0.17   | -   |                                                       | x |   |   |
| gson                 | 0.03    | 0.05    | +   | x                                                     |   |   |   |
| guava                | 0.00    | 0.04    | +   | x                                                     |   |   |   |
| guice                | 0.00    | -0.07   | -   | x                                                     |   |   |   |
| hadoop               | 0.00    | -0.30   | -   |                                                       | x |   |   |
| hazelcast            | 0.00    | 0.03    | +   | x                                                     |   |   |   |
| hbase                | 0.00    | -0.26   | -   |                                                       | x |   |   |
| hector               | 0.03    | -0.07   | -   | x                                                     |   |   |   |
| hibernate-orm        | 0.00    | -0.07   | -   | x                                                     |   |   |   |
| hibernate-search     | 0.00    | -0.58   | -   |                                                       |   |   | x |
| HMCL                 | 0.00    | -0.19   | -   |                                                       | x |   |   |

| project                   | p-value | d-value | +/- | <i>Effect size band (Cliff's <math>\delta</math>)</i> |   |   |   |
|---------------------------|---------|---------|-----|-------------------------------------------------------|---|---|---|
|                           |         |         |     | N                                                     | S | M | L |
| hutool                    | 0.00    | -0.14   | -   | x                                                     |   |   |   |
| Hystrix                   | 0.00    | -0.45   | -   |                                                       |   | x |   |
| iceberg                   | 0.00    | -0.23   | -   |                                                       | x |   |   |
| incubator-seata           | 0.00    | -0.04   | -   | x                                                     |   |   |   |
| intellij-community        | 0.00    | 0.34    | +   |                                                       |   | x |   |
| j2objc                    | 0.00    | -0.11   | -   | x                                                     |   |   |   |
| javaparser                | 0.50    | 0.01    | +   | x                                                     |   |   |   |
| jclouds                   | 0.00    | -0.20   | -   |                                                       | x |   |   |
| jedis                     | 0.00    | -0.12   | -   | x                                                     |   |   |   |
| jenkins                   | 0.00    | -0.14   | -   | x                                                     |   |   |   |
| jetty.project             | 0.00    | -0.12   | -   | x                                                     |   |   |   |
| jgit                      | 0.00    | 0.27    | +   |                                                       | x |   |   |
| jib                       | 0.32    | -0.02   | -   | x                                                     |   |   |   |
| jmeter                    | 0.00    | 0.15    | +   |                                                       | x |   |   |
| jna                       | 0.41    | -0.02   | -   | x                                                     |   |   |   |
| junit4                    | 0.00    | 0.32    | +   |                                                       | x |   |   |
| junit5                    | 0.00    | 0.08    | +   | x                                                     |   |   |   |
| kafka                     | 0.00    | -0.29   | -   |                                                       | x |   |   |
| lombok                    | 0.00    | 0.38    | +   |                                                       |   | x |   |
| lucene                    | 0.00    | -0.15   | -   | x                                                     |   |   |   |
| mage                      | 0.00    | -0.27   | -   |                                                       | x |   |   |
| metrics                   | 0.00    | -0.12   | -   | x                                                     |   |   |   |
| Mindustry                 | 0.80    | -0.02   | -   | x                                                     |   |   |   |
| MinecraftForge            | 0.00    | 0.63    | +   |                                                       |   |   | x |
| mockito                   | 0.21    | -0.02   | -   | x                                                     |   |   |   |
| mongo-java-driver         | 0.94    | 0.00    | +   | x                                                     |   |   |   |
| MPAndroidChart            | 0.02    | 0.42    | +   |                                                       |   | x |   |
| mybatis-plus              | 0.00    | 0.18    | +   |                                                       | x |   |   |
| nacos                     | 0.06    | -0.02   | -   | x                                                     |   |   |   |
| netty                     | 0.00    | -0.12   | -   | x                                                     |   |   |   |
| okhttp                    | 0.16    | 0.23    | +   |                                                       | x |   |   |
| openmrs-core              | 0.00    | -0.11   | -   | x                                                     |   |   |   |
| Paper                     | 0.35    | -0.02   | -   | x                                                     |   |   |   |
| pentaho-kettle            | 0.00    | 0.45    | +   |                                                       |   | x |   |
| plantuml                  | 0.00    | 0.46    | +   |                                                       |   | x |   |
| pmd                       | 0.00    | -0.35   | -   |                                                       |   | x |   |
| presto                    | 0.00    | -0.07   | -   | x                                                     |   |   |   |
| pulsar                    | 0.00    | -0.11   | -   | x                                                     |   |   |   |
| redisson                  | 0.00    | -0.16   | -   |                                                       | x |   |   |
| rocketmq                  | 0.00    | 0.10    | +   | x                                                     |   |   |   |
| RxJava                    | 0.00    | 0.25    | +   |                                                       | x |   |   |
| seatunnel                 | 0.00    | 0.12    | +   | x                                                     |   |   |   |
| shardingsphere            | 0.00    | 0.06    | +   | x                                                     |   |   |   |
| shardingsphere-elasticjob | 0.00    | -0.37   | -   |                                                       |   | x |   |
| shenyu                    | 0.02    | 0.03    | +   | x                                                     |   |   |   |
| Signal-Server             | 0.00    | -0.29   | -   |                                                       | x |   |   |
| skywalking                | 0.00    | -0.21   | -   |                                                       | x |   |   |
| SmartTube                 | 0.00    | 0.18    | +   |                                                       | x |   |   |
| solr                      | 0.00    | -0.16   | -   |                                                       | x |   |   |
| sonarqube                 | 0.00    | -0.30   | -   |                                                       | x |   |   |
| spring-boot               | 0.00    | -0.20   | -   |                                                       | x |   |   |

| project              | p-value | d-value | +/- | <i>Effect size band (Cliff's <math>\delta</math>)</i> |   |   |   |
|----------------------|---------|---------|-----|-------------------------------------------------------|---|---|---|
|                      |         |         |     | N                                                     | S | M | L |
| spring-cloud-alibaba | 0.30    | -0.05   | -   | x                                                     |   |   |   |
| spring-framework     | 0.00    | -0.22   | -   |                                                       | x |   |   |
| titan                | 0.00    | -0.15   | -   | x                                                     |   |   |   |
| tomcat               | 0.00    | 0.36    | +   |                                                       |   | x |   |
| vert.x               | 0.00    | -0.19   | -   |                                                       | x |   |   |
| visualvm             | 0.00    | -0.10   | -   | x                                                     |   |   |   |
| voldemort            | 0.00    | -0.28   | -   |                                                       | x |   |   |
| vraptor4             | 0.00    | -0.12   | -   | x                                                     |   |   |   |
| weka                 | 0.00    | 0.32    | +   |                                                       | x |   |   |
| wicket               | 0.00    | -0.05   | -   | x                                                     |   |   |   |
| wildfly              | 0.00    | 0.06    | +   | x                                                     |   |   |   |
| wiremock             | 0.00    | 0.04    | +   | x                                                     |   |   |   |
| xerces2-j            | 0.00    | 0.54    | +   |                                                       |   |   | x |
| xpipe                | 0.74    | -0.08   | -   | x                                                     |   |   |   |
| zxing                | 0.00    | 0.15    | +   |                                                       | x |   |   |
| all                  | 0.00    | -0.10   | -   | x                                                     |   |   |   |